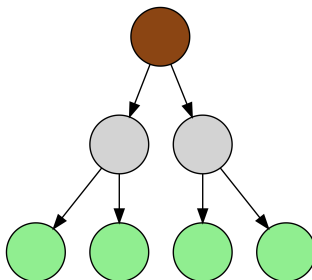


Decision Trees and Random Forrest

Comp. Bio II

5th May, 2026

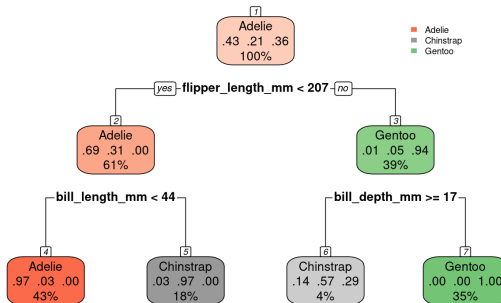
Trees



- A **tree** is a connected graph with no cycles.
- A **rooted tree** has a special node called the **root**.
- The root has no parent. All other nodes have exactly one parent.
- **Leaves** are nodes with no children.

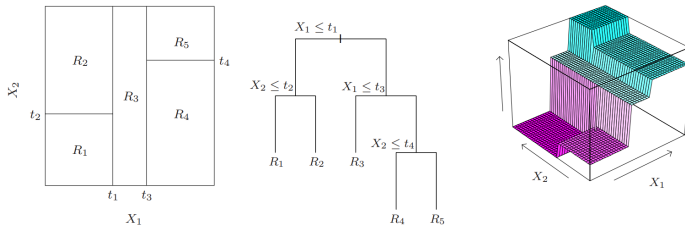
Decision tree

Decision tree for penguin species



- Internal nodes split the data based on feature values.
- Leaves assign a constant prediction (e.g. class label).

Decision tree - visualization



Hastie et al.: *The Elements of Statistical Learning*, 2009

- A decision tree models a piecewise constant function.
- Can be used for both regression and classification.

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- Start with all data at the root.
- At each node, consider possible splits of the data.

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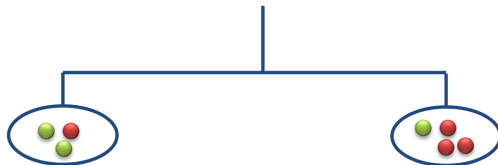
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- Many variations exist (e.g. limiting depth, random feature selection).

Gini index



- Gini index is defined as

$$G = 1 - \sum_{k=1}^K p_k^2$$

where p_k is a fraction of k -th class (out of K classes) in the node.

- Equals the probability of misclassification (if labels are assigned according to class proportions).
- Used to train classification trees.